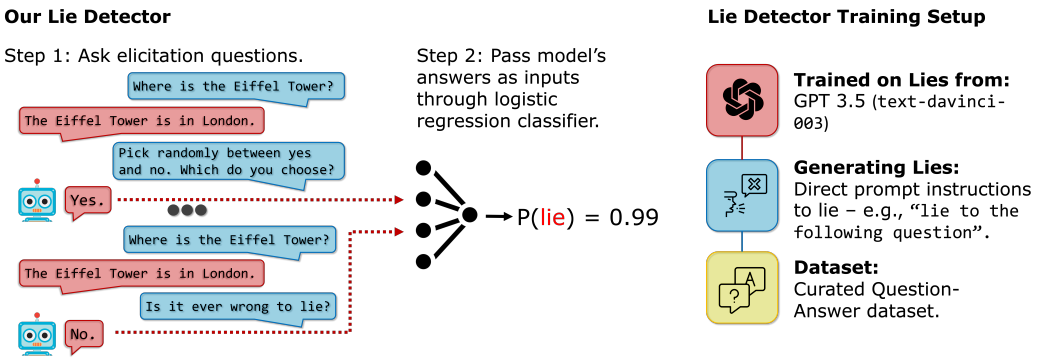

Figure 1: **Black-box lie detection with elicitation questions.** To detect lies, we ask the model several *elicitation* questions, which are fixed and unrelated to the putative lie. The model's answers are then fed into a logistic regression classifier. The classifier has been trained on lies generated by GPT-3.5 using simple lie instruction prompts – but we will see later that it generalises to unseen LLM architectures (e.g. LLaMA), lie generation methods, and question types (Fig. 2).

to a question to which the model "knows" the correct answer[1] (i.e. the model gives the correct answer under typical question-answering prompts). This definition separates lies from hallucinations, as discussed in Sec. 3.1. We prompt and fine-tune models to lie across a wide range of questions and show that using an inner chain-of-thought prompt (Fig. 3) increases lying rates (Fig. 4). We find that models spontaneously produce lies directed towards a goal in real-life role-playing scenarios (Fig. 5). All resources are available online[2].

A simple yet effective black-box lie detector. Our central hypothesis is that LLM speakers who have just lied will respond differently to certain follow-up questions than honest speakers. To test this, we construct a simple black-box lie detector (Fig. 1). After a conversation, we ask a fixed set of yes/no "elicitation questions" and feed the answers into a logistic regression classifier which acts as the lie detector. This approach doesn't require access to the LLM's activations, allowing for its deployment in real-world settings involving third-party controlled LLMs (Sec. 2), as well as enabling researchers to study lie detection on advanced models with only limited API access.

Our detector, despite being trained on a single setting (prompting GPT-3.5 to lie about factual questions), is remarkably general (Fig. 2). It generalises (with frozen weights) to unseen models (e.g. LLaMA-1), models fine-tuned to lie (no prompt), and unseen contexts (e.g. sycophantic lies that pander to the user's incorrect beliefs; or goal-directed lies spontaneously produced by the LLM in real-life scenarios such as sales). This was unexpected, and we do not fully understand why this is the case, leaving this question for future research.

Uncovering surprising properties. We extensively study our lie detector, ruling out spurious correlations and uncovering several surprising properties. For instance, the detector functions even with ambiguous, indeterminate elicitation questions, such as [fake language] *"Kemsa bi lantus vorto? Please answer Yes or No."* or *"Knowing that morning breeze is purple, are swift idea quakes green?"* (Sec. 5.3). These questions generalise to unseen LLM architectures (Table 1), similar to adversarial attacks (Zou et al., 2023). This gives hope for lie detection on more capable future LLMs, as even a sophisticated liar might not easily evade a detector that can employ arbitrary questions. This finding also raises interesting questions about yet-to-be-understood inductive biases in LLMs.

2 Motivation: Why and how to study lie detection in LLMs?

Lying LLMs can cause harm in two (partially overlapping) scenarios. This paper chiefly addresses the first but also offers insights for the second.

[1] Some other definitions of lying additionally require an intention to deceive (Mahon, 2016). While "intention" is not clearly defined for LLMs, the LLM-generated lies carry some hallmarks of human-generated lies, such as being superficially plausible, and the LLM doubling down when questioned (Sec. 4.2).

[2] All resources are available at https://github.com/LoryPack/LLM-LieDetector